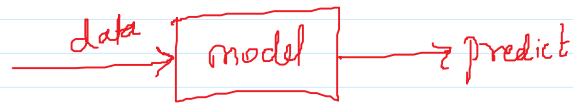


M.L $\rightarrow$ Subset of A.I

$\rightarrow$ Build the model



$\rightarrow$ Structured data is used in M.L

dataset

x	y
2	12.22
4	14.22
6	16.22
8	18.22
10	20.22
12	22.22
14	24.22
16	26.22
18	28.22
50	60.22

Train (75%)

Test (25%)

$y = x + 10.22$



learn the pattern
find the relation



$$\begin{aligned}
 \text{Error} &= \text{Actual} - \text{Pred} \\
 &= 60.22 - 79 \\
 &= \underline{\underline{-18.78}}
 \end{aligned}$$

70 : 30 $\rightarrow$ Train : Test

Index	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	y
0							
1							
2							
3							
4							
5							
6							
...							
...							
...							

X_{TRAIN}

y_{TRAIN}

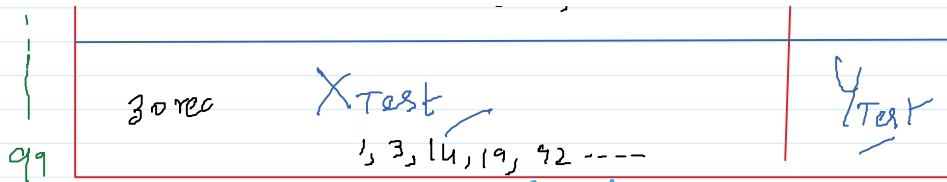
To rec $\rightarrow$ row index selecting Randomly
0, 15, 2, 4, 15, 28, 91, 72, ...

X_{TEST}

y_{TEST}

Training data 99
= 0, 3, 5, 15, 24, 28, 29
no. rec = 70

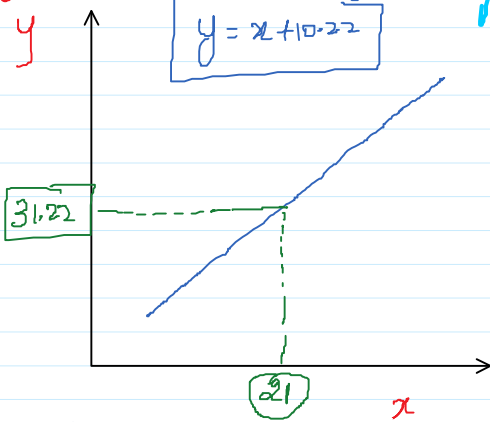
Test data



no rec = 10

$\}$ Test data
Unsupervised

Regression
 y

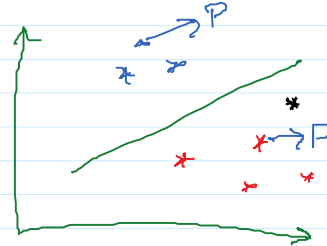


x	y
12	22.22

1) Kind of data y var $\Rightarrow$ Conti

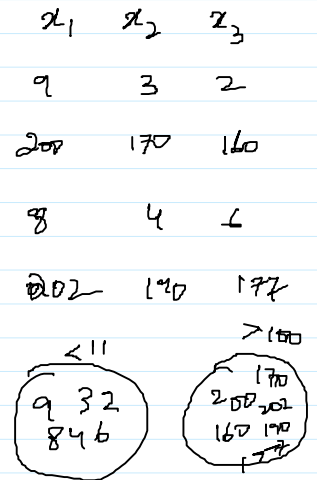
Supervised $m.l$ Classification

S	M	H	Result
90	80	75	F $\rightarrow 1$
35	20	18	F $\rightarrow 0$
90	81	83	P
25	18	33	F



* Kind $y \Rightarrow$ discrete data or categories

Clustering



* $x \neq 0$ Target

$m.l$

7:00 $\rightarrow$ Linear Reg

$\rightarrow$ Linear ipynb
 $\rightarrow$ Adres

Supervised $m.l$

i/p & o/p $\rightarrow$ model
(labelled data)

Regression
($y \rightarrow$ continuous)

Classifi
 $\rightarrow y \rightarrow$ discrete/categories

Unsupervised $m.l$

only i/p & No Target var
(Unlabelled data)

$\downarrow$
Clustering
 $\downarrow$
No Target

Linear Regression

$\rightarrow$ Supervised $m.l$

Linear Regression

→ Supervised M.L

→ labelled data

→ $x \rightarrow$ Features/ind
 $y \rightarrow$ Target var

→ always x & y should have a linear Reg *

→ there should not b/w i/p var (No Multicollinearity)

General eq: $y = m_1x_1 + m_2x_2 + \dots$
 $(-2)^2 = 4$

~~Residual~~ Residual Sum of Squares = $e_1^2 + e_2^2 + e_3^2 + e_4^2$
 $= (y_{a1} - y_{p1})^2 + (y_{a2} - y_{p2})^2 + \dots$

$$RSS = \sum_{i=1}^n (y_{ai} - y_{pi})^2$$

Validation

actual \ predicted

$\hat{y}$	y	
10	10.22	Train: 180
5	15.22	
4	14.22	
70	80.22	Test: 20
100	110.22	

(3) to find Best fit Line (Learn the patterns/Relation)

(1) Train data

LR model
(2)

$\rightarrow$ y predictions (y_p)

$$y_p = 81$$

$\uparrow$
 x Test data
(3)

$$x_{test} = 70$$

$\therefore$ Compare actual value (y_{test}) and $y_{prediction}$

$$Error = 80.22 - 81 = -1.22$$

Evaluate
the
model

R^2 Score

0.9 to 1 $\rightarrow$ Excellent

0.7 to 0.9 $\rightarrow$ Good

0.4 to 0.7 $\rightarrow$ Avg

0.4 to 0 $\rightarrow$ Weak

$< 0 \rightarrow$ Worst

$\frac{S+M+SOC+H+L+L}{600}$

% percentage
 $= 97\%$ FCD.

Logistic Regression Algorithm

$\rightarrow$ Supervised M.L

$\rightarrow$ labelled data

$\rightarrow$ Classification $\begin{cases} B_i \\ \text{Multi} \end{cases}$

$\rightarrow$ principle: Probability Based

$\rightarrow$ Target var $\rightarrow 0 \text{ or } 1 \rightarrow B_i$

$\rightarrow$ Log + Regression

$$z = m_1x_1 + m_2x_2 + \dots + c$$

Eg:-

D			
Sub1	Sub2	Sub3	y
			Pass
			Fail
			Fail
			Pass

Pass/Fail

Eg:- Percent = 85% Pass
= 30% Fail

Cut-off 35%

Sub $> 35 \rightarrow$ Pass

Sub $< 35 \rightarrow$ Fail

Pass $\rightarrow$ class 0 $\rightarrow$ true class

Fail $\rightarrow$ class, $\rightarrow$ -ve class

Balanced dataset $\xrightarrow{\text{Yes/No}}$ 100 rec Yes
100 rec No

Classification Unbalanced $\rightarrow$ 400 rec $\left. \begin{array}{l} \text{Yes class} \\ \text{100 rec} \end{array} \right\}$ No class } Model is biased towards Yes class

Technique $\rightarrow$